

I. QUESTION (ALGORITHM ANALYSIS)

Given the following function

```
1 int algorithm1(int N, int k){  
2     if (N == 0)  
3         return 0;  
4     int sum = 0;  
5     for (int i = 0; i < N; i++)  
6         sum++;  
7     for (int i = 0; i < N; i++)  
8         for (int j = i; j < N; j++)  
9             sum++;  
10    for (int i = 0; i < 4; i++)  
11        sum += algorithm(N / 2);  
12    return sum;  
13 }
```

What is the time complexity of algorithm1? Explain your result. Use the master theorem.

II. QUESTION (ALGORITHM ANALYSIS)

Given the following function

```
1 int algorithm1(int N){  
2   int i, j, k, sum = 0;  
3   for (i = 1; i <= N; i++)  
4     for (j = i; j <= N; j++)  
5       for (k = 1; k <= j - i; k++)  
6         sum++;  
7   return sum;  
8 }
```

What does the function algorithm1 return in terms of N?

III. QUESTION (LINKEDLIST)

Write the method

```
void doubleList()
```

which doubles each node in a doubly linked list, that is, after each node inserts a new node with the same value again. Write the method in C++ language.

IV. QUESTION (LINKEDLIST)

Write the method

```
void removeKth(int K)
```

which deletes K'th node from a singly linked list. Write the method in Java language.

V. QUESTION (STACK)

Implement the stack as a doubly linked list in C++ language. Your implementation must only contain a head node of type `DoubleNode*`, and three functions

`bool isEmpty(),`

`DoubleNode* pop()`

`void push(DoubleNode* newNode).`

VI. QUESTION (STACK)

Write a function in Java language which removes only the K bottom elements of the stack.

1 `void removeBottomK(int K)`

You are only allowed to use pop, push, isEmpty functions. You can use an external stack.